# Revision Within the Original Manuscript

**Controlling scope.** The revision retains the original title, authors, five main sections, methodology order, detector/learner families and three formal algorithms. Reviewer-requested material is inserted at the relevant locations within the submitted article.

**Comparison basis.** The confirmed original source is `07_Original_Archive/Submitted_V5_Confirmed/v5.tex`. Its page geometry, title block and citation style are retained. Figures 1–2 and their draw.io sources are unchanged original files.

| Original location | Revised location | Change and reason |
|---|---|---|
| 1. Introduction | 1. Introduction | Retains the problem and objective; adds explicit sequential/multilayer contributions and removes unsupported RL and performance claims. Reviewer point 1. |
| 2. LITERATURE REVIEW | 2. LITERATURE REVIEW | Reorganized into the four requested classes, with strengths, limitations, research gaps and two positioning tables. Forty cited references, including three relevant Annals papers. Reviewer points 1–2. |
| 3. METHODOLOGY | 3. METHODOLOGY | Same position and internal order: preprocessing; structure; anomaly detectors; supervised models; SAD structures; evaluation. |
| 3.1 PRE PROCESSING; Algorithm 1 | 3.1 PRE PROCESSING; Algorithm 1 | Retains Gini/NZV screening and SMOTE. Clarifies D1-only fitted transformations, named outcome removal and training-only resampling. |
| Structure; Figures 1–2 | Structure; Figures 1–2 | Restores exact original PNGs and editable draw.io sources. An adjacent note explains verified non-alert audits, training-only preprocessing and the unused final update. |
| ANOMALY DETECTION | ANOMALY DETECTION | All nine original detector families and EAD remain as individual descriptions. Adds precise settings and correct primary references; formalizes entropy gain. |
| 3.2 MACHINE LEARNING | 3.2 MACHINE LEARNING | Preserves LR, SVM, XGB, RF, NB, AdaBoost and ANN; restores the KNN paragraph already implied by the original eight-model code. Adds a settings table. |
| 3.3 SAD; Algorithm 2 | 3.3 SAD; Algorithm 2 | Retains Simple-A/B, vote thresholds, expert verification and updates. Explicitly verifies the PN sample and accumulates real labels; corrects batch-loop nesting. |
| 3.3 SAD; Algorithm 3 | 3.3 SAD; Algorithm 3 | Retains anomaly candidates $Y_1$, EAD gate $C_1$, shortlist $F_1$ and supervised update cycle. Uses candidate-local entropy and the same review correction. |
| 3.4 EVALUATION | 3.4 EVALUATION | Retains accuracy, recall and F1 formulas; defines fraud as positive and adds precision, specificity, balanced accuracy, ROC–AUC, AP and workload. |
| 4. Real Data Application | 4. Real Data Application | Retains Carclaims and its descriptive frequency table; replaces invalid/ambiguous old performance exports with auditable corrected results. |

| Original location | Revised location | Change and reason |
|---|---|---|
| Algorithm specifications | 4.2 Algorithm specifications | Retains 50/30/10/10 partitions and original architecture/model sets; states ten seeds, fixed settings and necessary implementation corrections. |
| No second-benchmark validation | 4.1 and 4.3–4.6 | Adds Insurance Claims under the same SAD settings; cites the researcher-confirmed mexwell/Kaggle source. Original insurer provenance remains unverified. Reviewer point 3. |
| Original model comparisons | 4.4 Classifier comparisons | Adds balanced forest and adapted PSO-XGBoost alongside eight component classifiers, using a common acquired-label policy. Reviewer point 3. |
| Original sensitivity discussion | 4.5 Ablation and sensitivity | Adds frozen-model sequential ablation and separates entropy placement comparisons; reports two genuinely rerun joint threshold pairs. Reviewer point 3. |
| 5. CONCLUSION | 5. CONCLUSION | Same final section. Interprets actual results, limits claims of superiority and includes limitations/future work here, without a new main Discussion section. |
| R research archive | Full preserved original archive | Original R code/exports remain unchanged. Models/results remain Python. Executed R/ggplot2 draws all eight empirical figures from those results. |
| Plot presentation | Figures 3–10 | Restores gray panels, facets, colours, line types and symbols. Workload is plotted by partition. The old exhaustive threshold grid is not claimed or fabricated. |
| Computational assistance | 3.4 | Briefly acknowledges manuscript and analysis/plotting-script assistance, with author responsibility. |
| Final editorial review | Throughout; 4.6 | Edits language using the supplied similarity report; improves figure labels and pagination; adds public data/code availability through the authors' GitHub repository. No model or input-data changes. |

**Numerical corrections.** The original exports had positive-class, leakage, indexing, entropy and resampling discrepancies. The revised figures therefore use the corrected, previously executed results. This final editorial pass changes no input data, core model modules or numerical result files.

**Reading changes.** Blue marks substantive additions and corrections. Minor language/citation edits are not individually colored. The unified source diff against the confirmed submission records additions and deletions.

**Final editorial pass.** Language, captions, figure readability and pagination were reviewed. The second dataset source and data/code availability are explicit. Author names remain; the authors' GitHub repository is linked in the manuscript and response letter.

**Submission files.** Use the inner source-only journal archive for the editable manuscript and reviewer response. It contains no PDFs, including figure PDFs. The complete package additionally contains compiled reading copies, the marked version, the Persian report, data, code, outputs, vector graphics and historical files.